Equation Branches in the LHC Black-Hole Safety Literature

Public mechanism map from source-derived artifacts

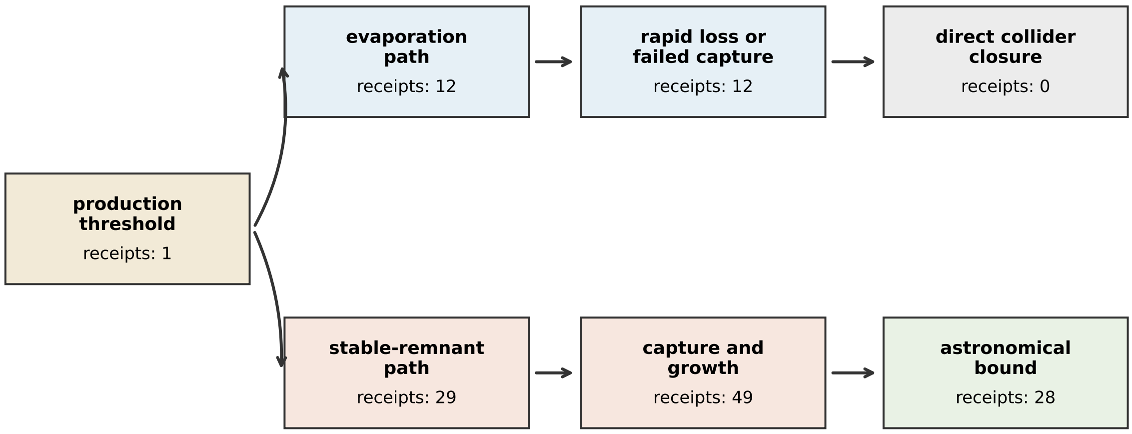
Abstract

The LHC microscopic-black-hole safety case is a branch problem in physics. A dangerous branch would require production, survival, capture, growth and failure of the astronomical bound. A safe branch can close through evaporation, failed capture, slow accretion or compact-object survival constraints. We build a public mechanism map of this literature from source-derived artifacts. The selected run contains 492 arXiv sources, 377 extracted claim labels and 1,408 equation witnesses. Formula filters reduce these witnesses to 204 usable mechanism nodes and 35 evidence-grade case nodes. The result is a branch map: 0 direct formula-clean LHC-safety mechanisms, 1 collider-threshold hook and 28 astrophysical black-hole analogues. The main result is a mechanism-transfer problem: production, evaporation, capture, accretion and compact-object survival must be connected by explicit equations before an adjacent astrophysical argument can support a collider safety conclusion.

1 The safety case is a branch structure

The LHC black-hole debate turned on a branch structure in physics: whether microscopic black holes could be produced, whether they evaporate, whether a stable remnant could be captured, how fast it could accrete matter, and whether astronomical observations already exclude the dangerous branch. Those questions appear in the published safety literature and its critics [1, 2, 3, 4, 5, 6].

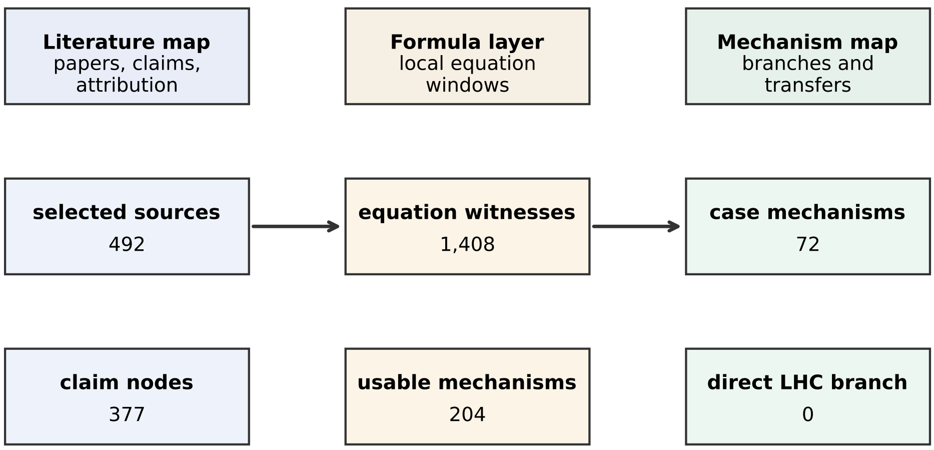
For a non-specialist, the essential point is simple. A dangerous scenario needs every step of one chain to work: the collider must make the object, the object must survive, it must be trapped, it must grow fast enough, and astronomical objects must fail to rule out the same growth mechanism. A safety argument can close the chain at several places: no production, rapid evaporation, failed capture, slow growth, or contradiction with observed compact-object survival. The evidence consists of equation-supported branch tests distributed across the source set.



The public run concentrates evidence in the adjacent astrophysical branch.

Figure 1: Physical branch structure of the LHC black-hole safety case. Counts are evidence-grade receipts retained by the public run. The populated route is mainly the adjacent astrophysical branch; the direct collider branch remains empty under the strict formula filter.

The public repository first builds a literature map of papers, claim labels and source-to-claim edges. It then builds an equation mechanism map from local formula windows, formula-quality filters, route labels and cross-source analogues. The literature map locates the discussion. The equation map asks which physical branches are populated by inspectable formula receipts.



The source map locates the discussion. The mechanism map locates the physical branches carried by equations.

Figure 2: Source and equation maps extracted from the same literature. The literature map records attribution and claim labels. The equation mechanism map records whether local formulas populate the physical branches of the argument.

2 From formulas to mechanism receipts

An equation witness is a local formula window extracted from a source. It becomes a usable mechanism node only when it survives formula-quality filters and activates at least one route. The six route labels are broad physical operations:

$$\begin{array}{ll} \text{transport or flow} & \partial_t q + \nabla \cdot J = S, \\ \text{constraint or closure} & C(q, J, \lambda) = 0, \\ \text{spectral or operator} & Lq = \lambda q, \\ \text{boundary or weak form} & \int_\Omega \phi L q = \int_\Omega \phi f, \quad Bq = b, \\ \text{commutator or residual} & [A, B] = AB - BA, \\ \text{ordered protocol} & x_{n+1} = \Phi(x_n, u_n). \end{array}$$

The labels act as formula filters. They identify the role played by an equation inside a possible derivation while leaving the physics in the source-local receipt.

This is the practical definition of evidence used in the report. A paragraph that says a black hole is safe or dangerous is a claim. A formula such as a mass-loss rate, an accretion rate, a threshold condition or a compact-object survival bound is a mechanism receipt. The receipt can be inspected because it states what has to change, what quantity is conserved or constrained, and what observation would matter.

For the LHC case, the relevant branch can be written schematically as

$$\text{production} \to \text{evaporation or stability} \to \text{capture} \to \text{growth} \to \text{astrophysical constraint.}$$

This is the mechanism structure behind the safety argument. The source map records where the discussion occurs. The equation mechanism map asks which parts of the branch are represented by formulas in the selected corpus.

3 What the run found

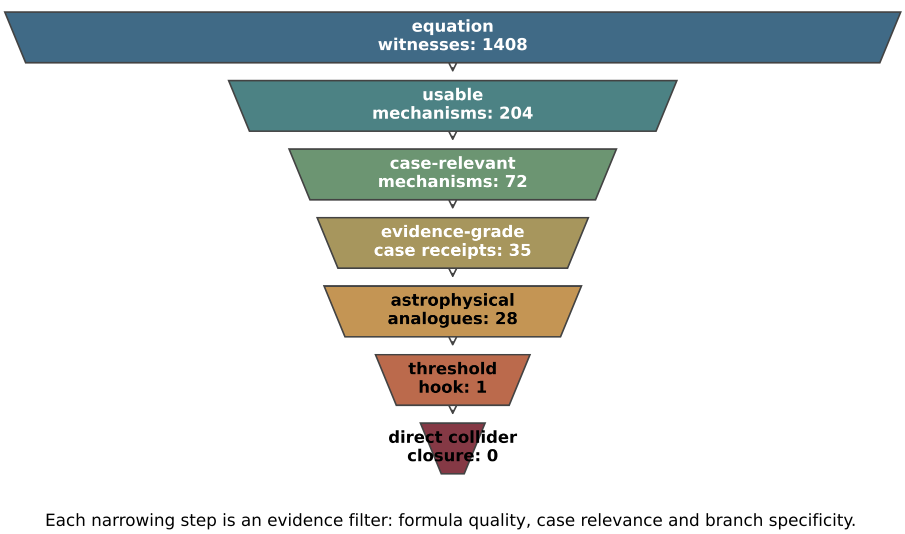


Figure 3: Evidence funnel under the public formula filter. Most equation windows are kept as trace material or rejected. The retained layer is smaller and physically more informative than a word-level source map.

The selected run starts with 1,408 equation witnesses. The formula filter retains 204 usable mechanism nodes and 35 evidence-grade case nodes. The direct LHC-safety branch has no formula-clean node in this run. The adjacent branch is populated: the graph finds 28 astrophysical black-hole analogues and 1 collider-threshold hook.

The result is a map of what the selected corpus can prove locally. It proves that the run contains many formula-clean black-hole mechanisms relevant to growth, capture and compact-object bounds. It also shows where the selected corpus is thin: direct collider-safety formulas are absent from the populated branch. The safety argument therefore depends on mechanism translation: adjacent astrophysical mechanisms must be carried into the collider setting with their assumptions visible.

Published safety analyses use cosmic-ray and astronomical constraints in addition to direct collider reasoning [1, 2, 4]. In this public run, the source-local formula-clean receipts sit mainly in adjacent black-hole physics. The graph therefore makes the collider translation problem explicit.

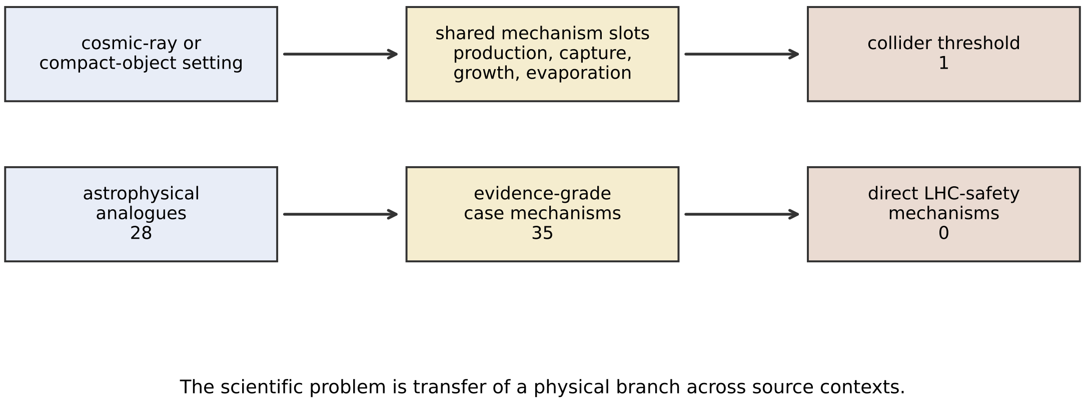


Figure 4: Mechanism translation exposed by the graph. The populated side of the run is adjacent astrophysical black-hole physics. The collider side contributes a threshold hook; direct source-local safety mechanisms are absent from the formula-clean retained layer.

4 The physical branches

The production branch asks whether the collider reaches the threshold or event condition assumed by a microscopic-black-hole model. A minimal form is

$$\sqrt{s} > M_D.$$

The selected public run found one threshold or event-selection hook. That is a small branch, so the report treats it as a hook, not as a closed collider derivation.

The evaporation branch is different. Hawking radiation supplies the standard mass-loss route for microscopic black holes [7]. A receipt in this branch should support a sign or timescale relation such as

$$\frac{dM}{dt} < 0, \qquad \tau_{\mathrm{evap}} \ll \tau_{\mathrm{capture}}.$$

The stable branch reverses the problem. If a black hole does not evaporate before capture, the relevant equations become stopping, capture and accretion:

$$\frac{dM}{dt} = \rho\,\sigma(M)\,v.$$

Astronomical survival then becomes a constraint. If the same mechanism would destroy white dwarfs or neutron stars on short timescales, observed survival of those systems constrains the dangerous branch. This is why the safety argument is best read as a transfer of a mechanism from collider production to compact-object capture and growth.

These equations are schematic, but they show what the graph is looking for. A threshold formula belongs to production. A negative mass-change or short lifetime belongs to evaporation. A positive growth law belongs to the stable-remnant branch. A compact-object lifetime bound belongs to the astronomical test. The graph is useful because it keeps these roles separate instead of mixing them into one textual controversy.

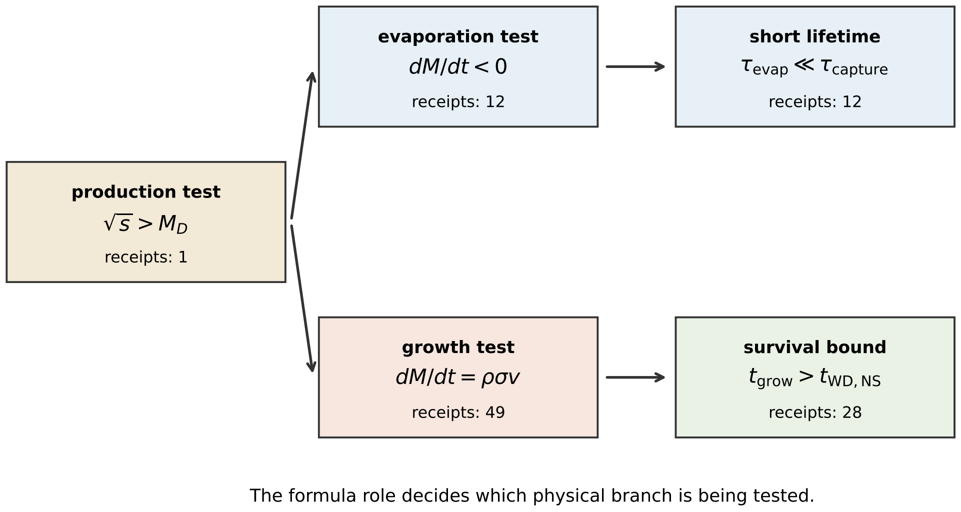


Figure 5: Equation tests behind the physical branch map. The same source set is sorted by formula role: production threshold, evaporation timescale, growth rate and astronomical survival bound.

5 Route structure and sparse-attention result

The mechanism routes retained by the graph are concentrated in closure, operator structure and transport:

Quantity	Count
constraint closure	165
spectral operator	160
transport flow	95
boundary weak form	28
discrete protocol	24
commutator incompatibility	12

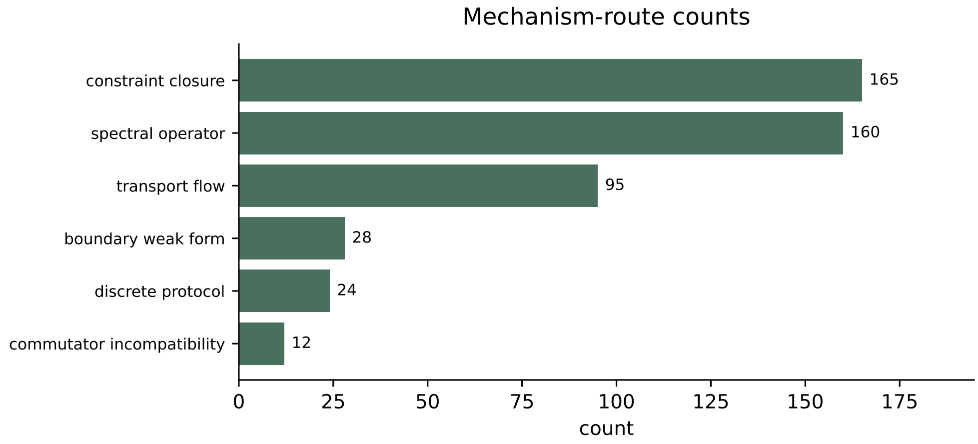


Figure 6: Mechanism-route counts among usable nodes. The retained layer is dominated by closure, operator structure and transport, the routes needed for growth, evaporation and capture arguments.

The route co-activation calculation supports the same interpretation. In the evidence-grade case layer, the largest route counts are spectral operator (34), constraint closure (33), transport flow (25), boundary weak form (4). Astrophysical analogues distribute across spectral operator (0.35), constraint closure (0.34), transport flow (0.26), which is the expected pattern for mass, capture and growth mechanisms. The strongest co-activations are constraint closure + spectral operator (32), spectral operator + transport flow (25), constraint closure + transport flow (23). This route-level calculation marks shared equation operations across source contexts. Physical equivalence still depends on the assumptions carried by each source.

In plain terms, the retained formulas show repeated structure. They combine three operations: an operator or rate law, a closure or constraint, and a transport or growth term. That pattern is exactly what a black-hole safety branch requires. A growth scenario needs a rate law and a constraint on available matter. An evaporation scenario needs a rate law and a timescale. An astrophysical bound needs a constraint that connects the same mechanism to observed survival.

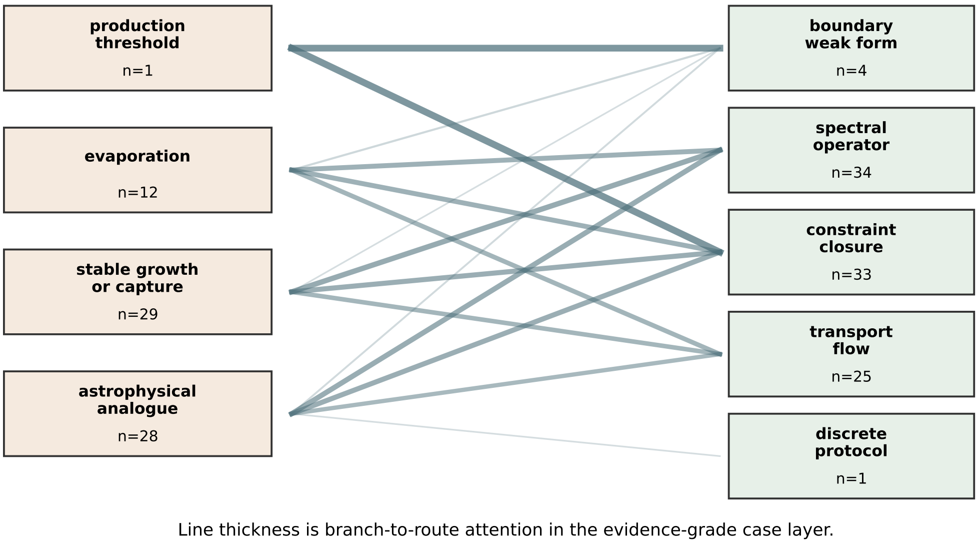


Figure 7: Branch-to-route graph in the evidence-grade case layer. Branches are on the left, equation routes are on the right, and line thickness gives route attention. The dominant branch connections run through spectral/operator, closure and transport operations.

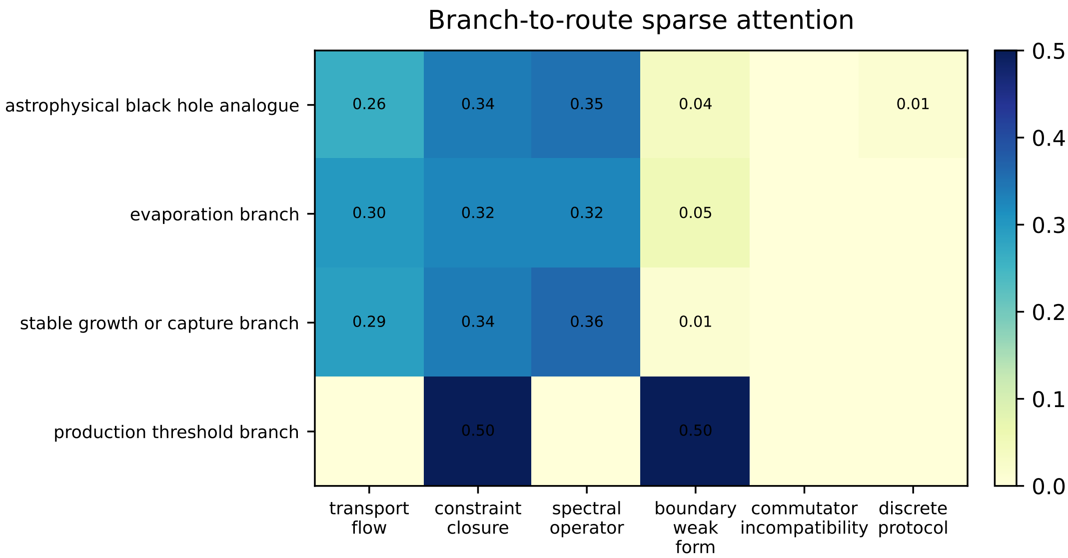


Figure 8: Branch-to-route sparse attention. Adjacent astrophysical mechanisms concentrate in operator, closure and transport routes; the production-threshold branch is separate and small.

6 Source map result

The source map has 869 nodes and 377 source-to-claim edges. Its claim labels are:

Quantity	Count
astrophysical claim	355
risk claim	20
safety claim	2

Table 1: Representative mechanism receipts retained by the formula filter.

Source	Routes	Branch	Formula witness
0904.0230	constraint closure, boundary weak form	production threshold branch	`|y_{\gamma \gamma}| \leq 1`
0807.3458	transport flow, constraint closure, spectral operator	astrophysical black hole analogue, evaporation branch, stable growth or capture branch	`3 \times 10^{-13}~\mathrm{M}_{\odot}~\mathrm{yr}^{-1} \lesssim \langle \dot{M}_{\mathrm{long}} \rangle \lesssim 1 \times 10^{-10}~\mathrm{M}_{\o...`
0807.3458	transport flow, constraint closure, spectral operator	astrophysical black hole analogue, evaporation branch, stable growth or capture branch	`4 \times 10^{-14}~\mathrm{M}_{\odot}~\mathrm{yr}^{-1} \lesssim \langle \dot{M}_{\mathrm{long}} \rangle \lesssim 2 \times 10^{-11}~\mathrm{M}_{\o...`
astro-ph0212297	constraint closure, transport flow, spectral operator	astrophysical black hole analogue, evaporation branch, stable growth or capture branch	`B_c\simeq 10^{16}\mbox{G}\left({7M _\odot}/{M_H}\right) \left({6M_H}/{R}\right)^2\left({M_T}/{0.03M_H}\right)^{1/2}`

This layer is still useful. It records which sources belong to the safety, risk, astrophysical-bound or collider-threshold discussion. It is also the right layer for dates, author identity and citation provenance. Its limit is categorical: source-to-claim edges leave the evaporation, capture and accretion branches unresolved until the equation layer is inspected.

The source map therefore plays a supporting role. It tells the reader where the discussion sits and which papers are involved. Closure of a mass-growth branch requires the local formulas and their physical roles.

7 Mechanism receipts

The report exposes local formula windows that can be checked against the source papers. Table 1 shows a compact sample; the full receipt set remains in `equation_mechanism_graph.json` and `equation_mechanism_graph.md`. The table should be read as an evidence ledger. Each formula is a reason why a branch was counted: a mass accretion rate, a bound on black-hole mass change, a compact-object scale or a threshold-like condition.

8 Evidence register

Claim	Evidence	Interpretation
Literature map	869 nodes and 377 source-to-claim edges.	This layer identifies documents, claim labels and attribution. Branch closure is tested in the equation mechanism map.
Equation mechanism map	1408 equation witnesses reduced to 204 usable mechanism nodes.	The evidence filter removes prose fragments and word-only matches, leaving equation windows that can be inspected.
Case split	0 direct LHC-safety mechanisms, 1 threshold hook and 28 astrophysical analogues.	The selected corpus populates adjacent black-hole mechanisms more strongly than direct collider-safety derivations.
Route content	constraint closure=165, spectral operator=160, transport flow=95	The dominant equation roles are closure, operator structure and transport, which are the routes needed for growth or evaporation branches.
Sparse check	In the evidence-grade case layer, the largest route counts are spectral operator (34), constraint closure (33), transport flow (25), boundary weak form (4). Astrophysical analogues distribute across spectral operator (0.35), constraint closure (0.34), transport flow (0.26), which is the expected pattern for mass, capture and growth mechanisms. The strongest co-activations are constraint closure + spectral operator (32), spectral operator + transport flow (25), constraint closure + transport flow (23).	This calculation uses the public graph artifacts and supports the branch interpretation.

9 Formula-quality boundary

The graph carries its own failure modes. Formula-quality labels in the public run include:

Quantity	Count
prose or latex artifact	947
no relation operator	689
word heavy formula	372
formula core	299
sentence boundary inside formula	272
long prose fragment	6

These counts explain why the evidence funnel is steep. Many local windows are prose, bibliography, weak formula fragments or equations without a case-relevant mechanism. The formula filter is deliberately conservative: noisy material remains available in the source-derived artifacts, while formula-clean case windows are promoted to mechanism receipts.

10 Conclusion

The public run demonstrates a practical difference between two ways of reading scientific papers. A source map reconstructs attribution and claim labels. An equation mechanism map asks whether the equations needed by the argument are present and how they connect. For the LHC black-hole case, that second graph exposes the main structure of the evidence: a sparse direct collider branch, a small threshold hook, and a populated set of adjacent astrophysical mechanisms. The scientific question becomes a controlled transfer problem between physical branches.

References

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[6] R. Casadio, S. Fabi and B. Harms, "Possibility of catastrophic black hole growth in the warped brane-world scenario at the LHC," arXiv:0901.2948 (2009).

[7] S. W. Hawking, "Particle creation by black holes," Communications in Mathematical Physics 43, 199–220 (1975).